

Fluid Mechanics

Boundary layers

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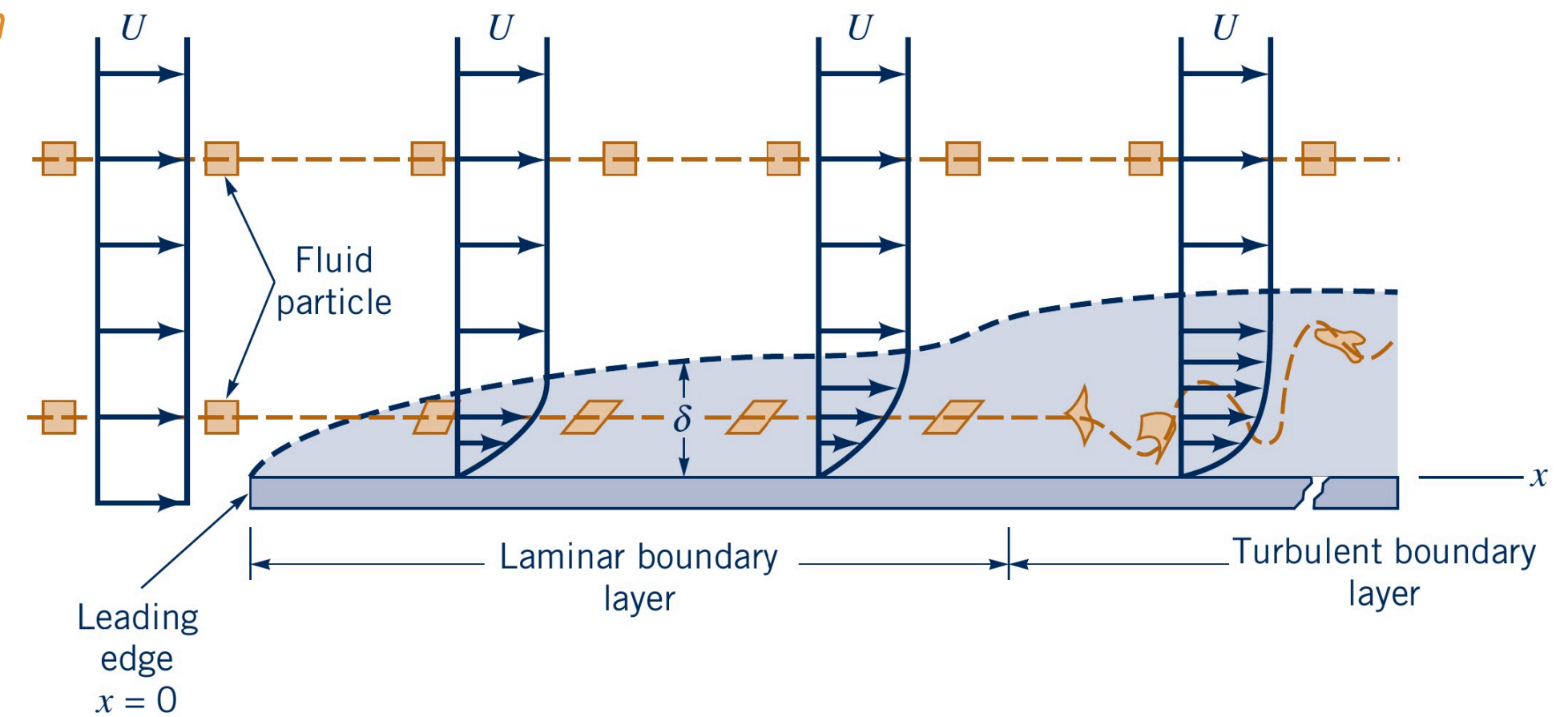
Boundary Layer Structure

Boundary layer (BL) structure and development for *parallel flow over an infinite flat plate*

- Only fluid in BL "feels" the plate
- $Re_x = \rho U x / \mu$ (bulk flow definition)

Laminar BL: Analytical expressions from *Blasius BL solution*

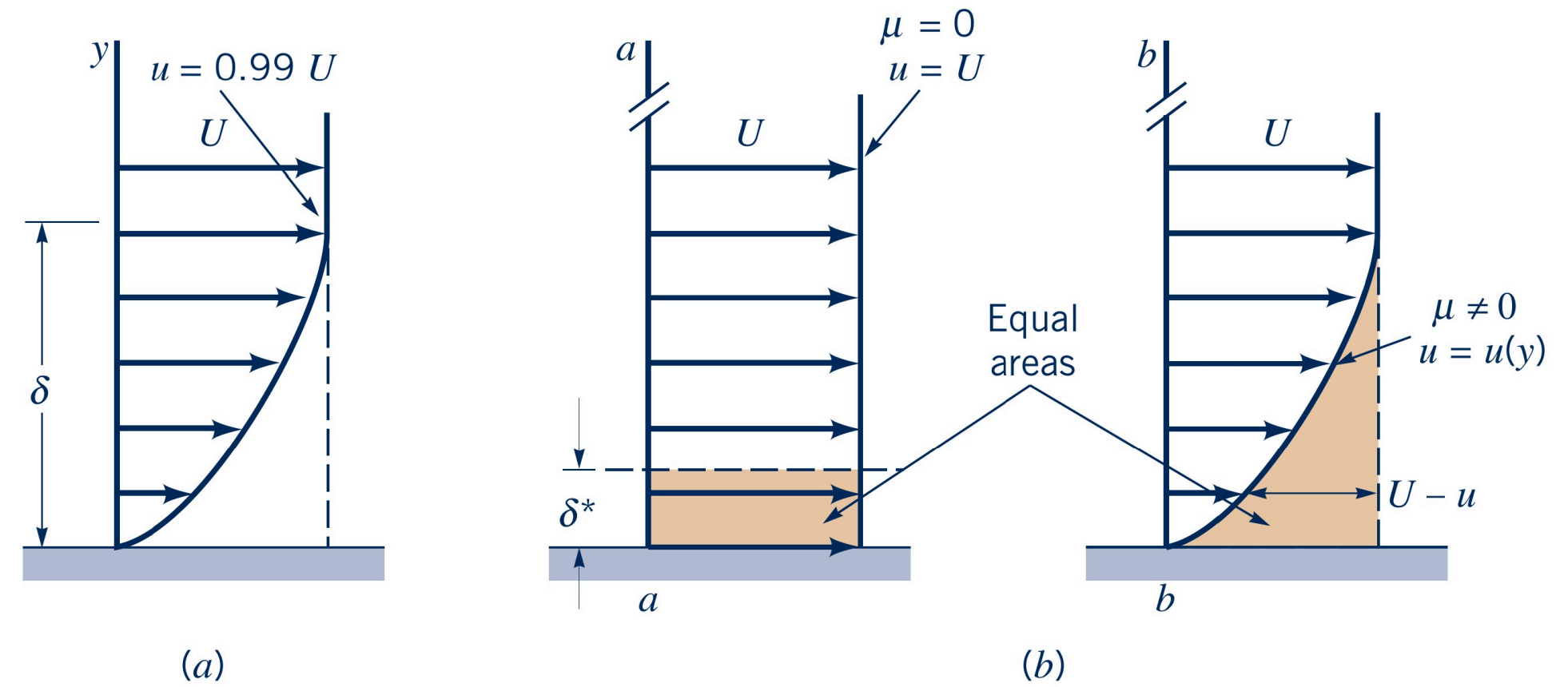
Turbulent BL: Mostly *empirical* expressions



Boundary Layer Thickness

BL thickness definitions:

- *Standard BL thickness* $\delta_{99} = \delta$
- *BL displacement thickness* δ^*
- *BL momentum thickness* Θ



Parallel flow to an (almost) infinite flat plate

Excellent experiment showing **turbulent BL** triggered by trip wire in this classical flow case

- Note: *Friction Reynolds* number $\text{Re}_\tau = \frac{\rho \sqrt{\tau_w / \rho} \delta}{\mu}$ (local) rather than *bulk Reynolds number* $\text{Re}_x = \frac{\rho U x}{\mu}$ is used in video

Spatially developing turbulent boundary layer on a flat plate
fyfluidynamics



Watch on

Turbulent Shear Stress

Recall how the instantaneous (horizontal) velocity u could be decomposed into a time-averaged and fluctuating part (*Reynolds decomposition*):

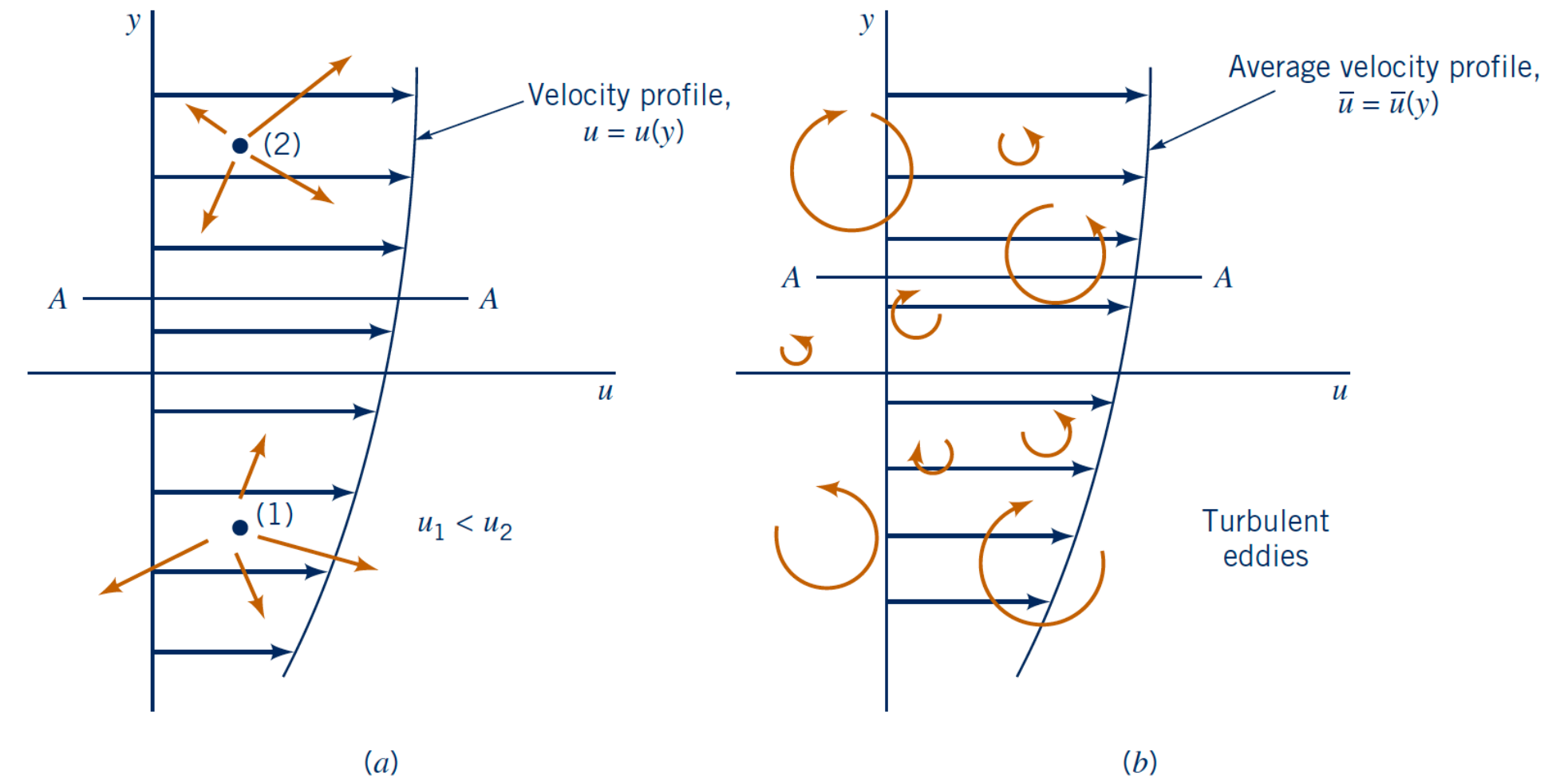
$$u(x, y, z, t) = \bar{u}(x, y, z) + u'(x, y, z, t)$$

- Tempting to apply our Newtonian shear stress definition directly to $\bar{u}$ in turbulent BL
- However, $\tau \neq \mu \frac{d\bar{u}}{dy}$ (with turbulence)

Let us review the underlying physics:

- **Laminar BL**
 - Molecular diffusion
 - x -momentum transfer in y due to $\frac{du}{dy} \neq 0 \rightarrow$ Shear stress
- **Turbulent BL**
 - Finite size 3-D eddies
 - Increasing momentum transfer
 - Extra term in shear stress:

$$\tau = \mu \frac{d\bar{u}}{dy} - \rho \overline{u'v'} = \tau_{lam} + \tau_{turb}$$



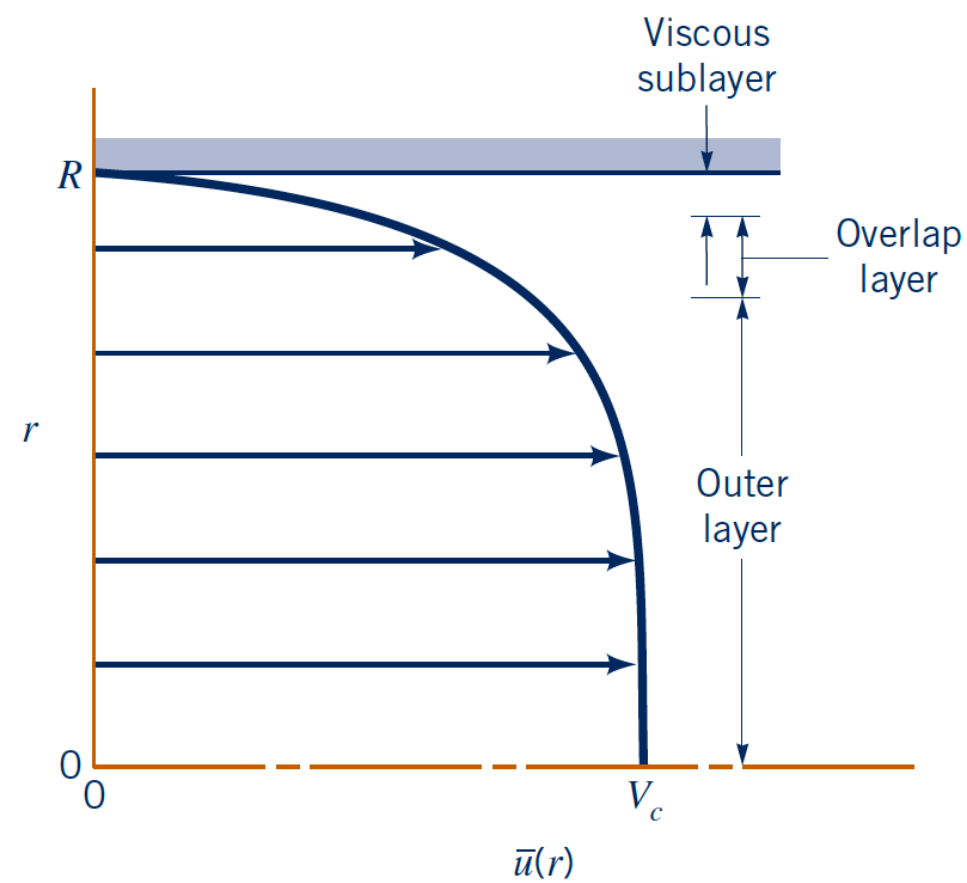
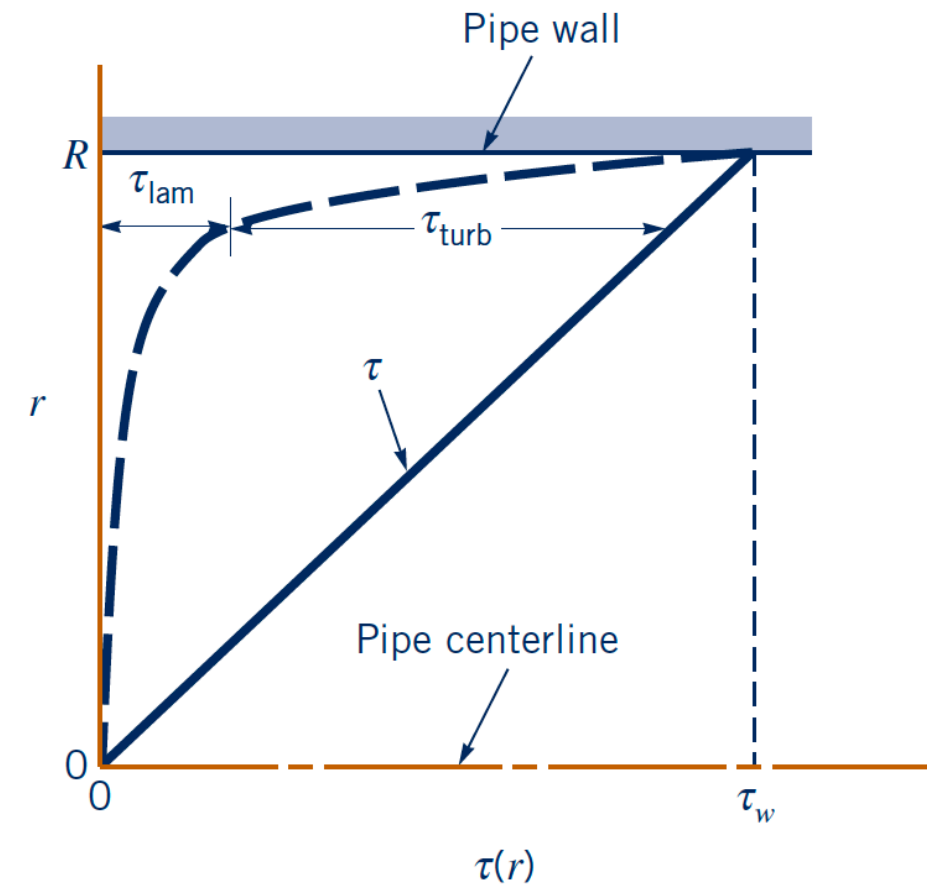
Reynolds Stresses

The additional contribution to shear stress from fluctuations $\tau_{turb} = -\overline{\rho u'v'}$ has units of Pa and is termed *Reynolds stresses*.

- These stresses are flow dependent and difficult to describe
 - *Fluctuating* velocities influence the *mean* flow
 - **Boussinesq hypothesis:**

$$\tau_{turb} = -\overline{\rho u'v'} = \mu_t \frac{d\bar{u}}{dy}$$

- Shifts the problem from Reynolds stresses to μ_t (flow dependent scalar)
- The basis of widely used RANS and URANS turbulence models
- Typical ratio of τ_{lam} and τ_{turb} in turbulent BL varies with distance to the wall (here for a pipe flow)



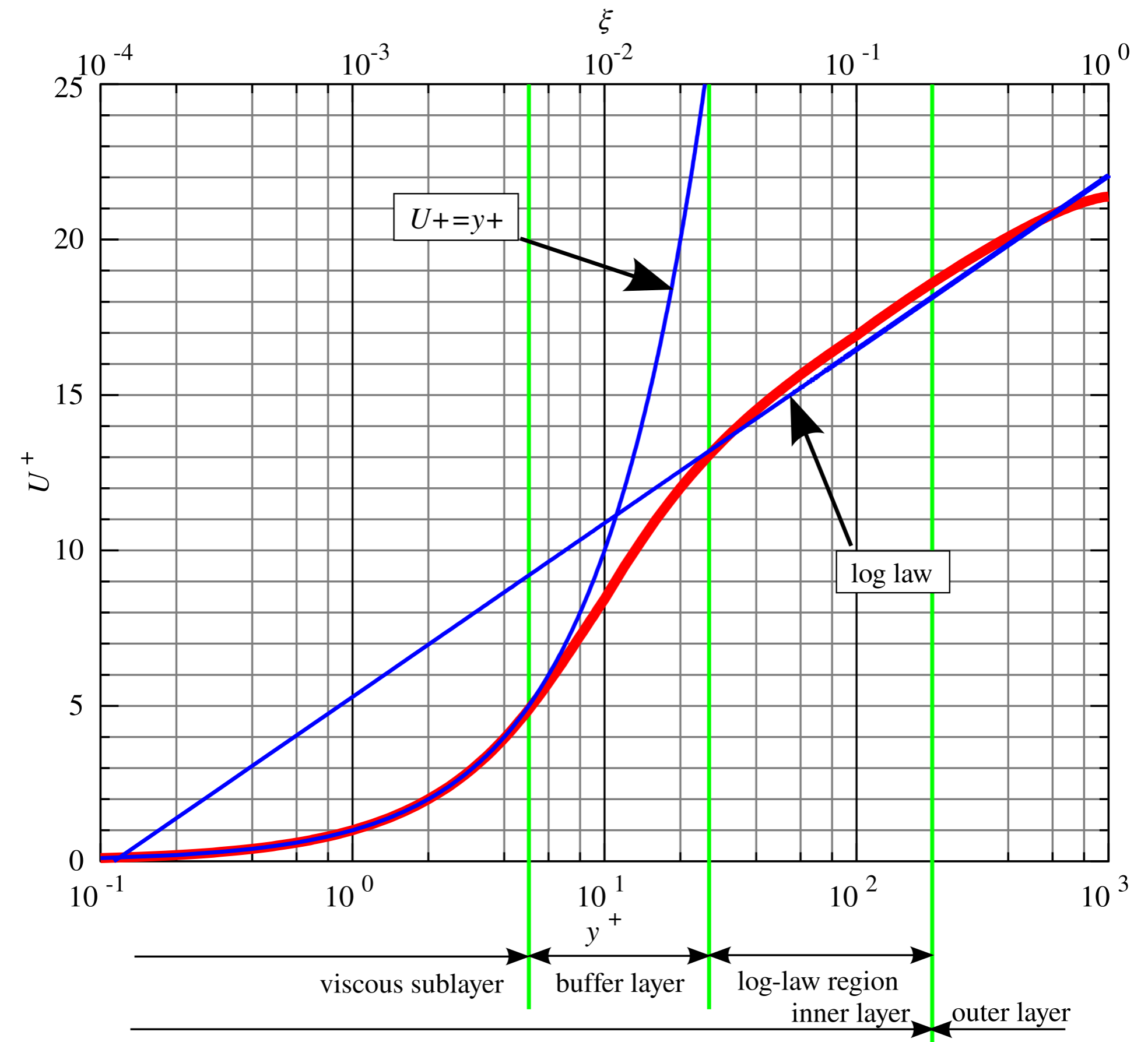
Turbulent Velocity Profile

Turbulent BL: *Three regions* with distinct physics given from distance to wall

- Dimensionless y coordinate: $y^+ = yu^*/\nu$ with $u^* = \sqrt{\tau_w/\rho}$
- Dimensionless velocity: $u^+ = \bar{u}/u^*$
- *Viscous sublayer* ($y^+ < 5$)
 - Viscous forces dominant over Reynolds stresses
 - Smooth wall: $u^+ = y^+$
- *Buffer layer* ($5 < y^+ < 30$)
 - Both viscous sublayer and log law important (transition function)
- *Log-law layer* ($30 < y^+ < 200$)
 - Velocity varies as the logarithm of y
 - Experimentally determined (law of the wall):

$$u^+ = \frac{1}{\kappa} \ln(y^+) + C^+$$

- Where κ is the von Kármán Constant: $\kappa = 0.41$ and $C^+ = 5$ (for smooth walls and flow over infinite flat plate)
- Viscous sublayer extremely thin
- **Roughness** of the wall can enter viscous sublayer $\rightarrow$ changed $u(y)$

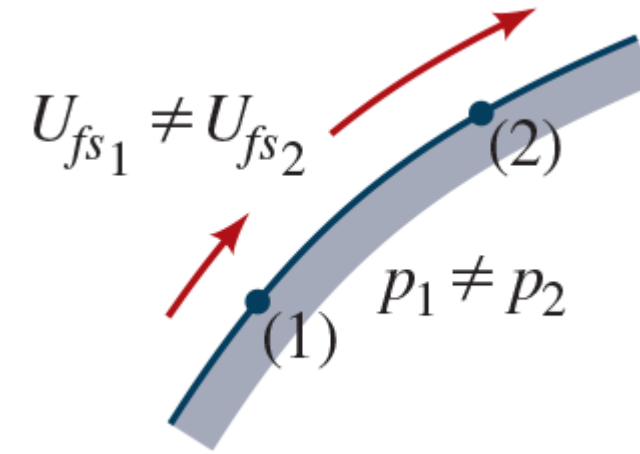


Pressure Gradients and Boundary Layers

Flat plate: uniform outer flow does not yield a pressure gradient (streamlines negligibly deformed).

Bluff or curved body: changing outer streamlines cause a pressure gradient along the surface.

- *Favorable:* pressure decreases and the outer flow accelerates.
- *Adverse:* pressure increases and the outer flow decelerates.



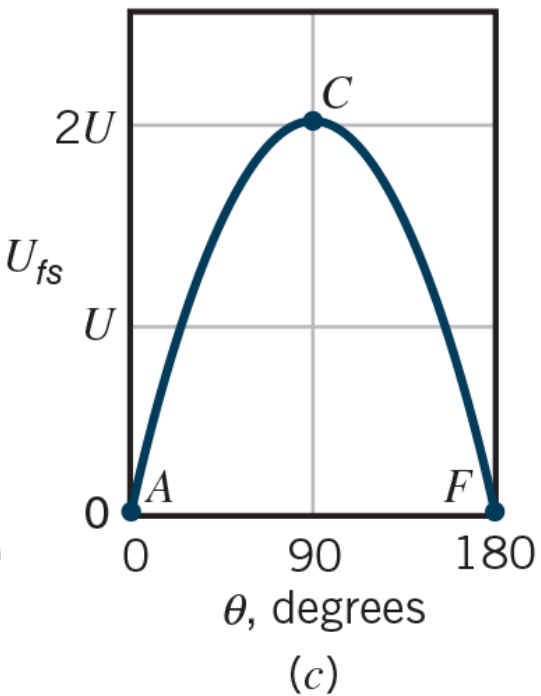
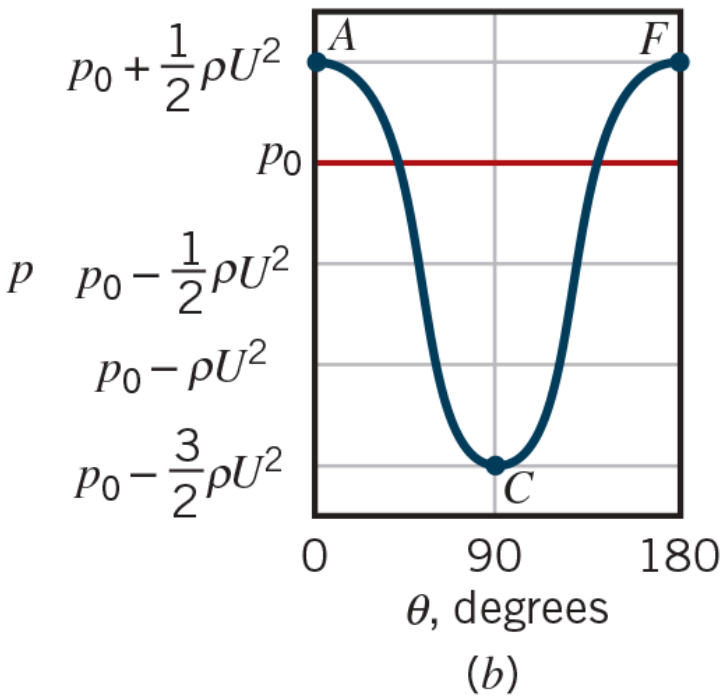
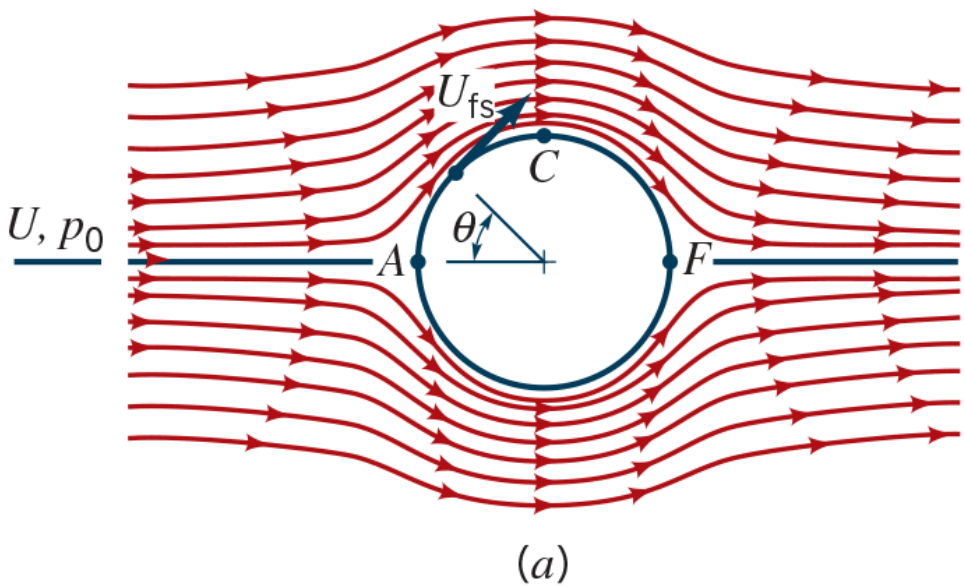
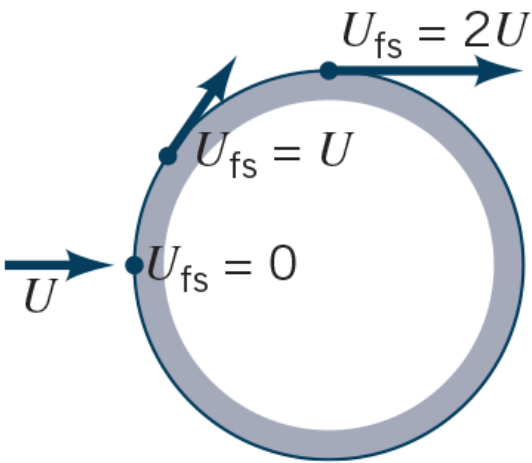
Inviscid Flow Around a Circular Cylinder

Inviscid theory can give a solution for the pressure distribution and drag on a circular cylinder.

- The free-stream speed is zero at the front and rear stagnation points, and maximum at the top and bottom of the cylinder.
- The inviscid pressure distribution is symmetric from front to rear.
- With no viscosity there is no wall shear, and the symmetric pressure forces cancel:

$$\mathcal{D} = 0$$

- *Would we expect this from common experience?*



Viscous Flow Around a Circular Cylinder and d'Alembert's Paradox

Adding viscosity introduces viscous effects and a finite drag can be measured (as we would expect).

However, what happens when μ approach zero?

d'Alembert's paradox unfolds:

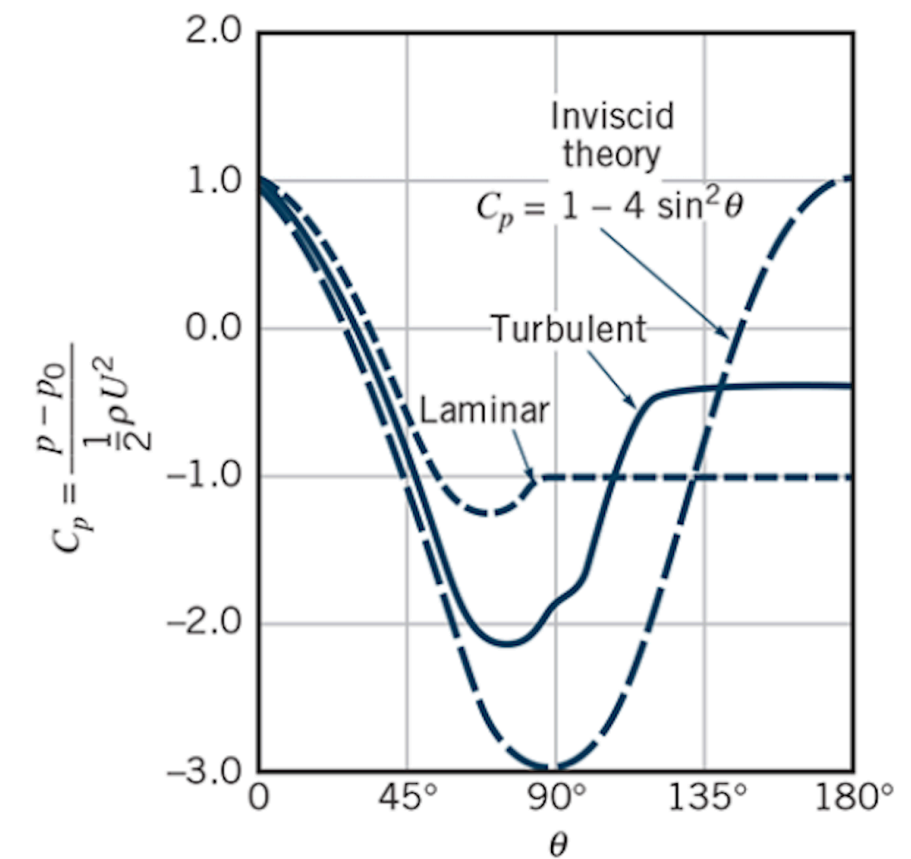
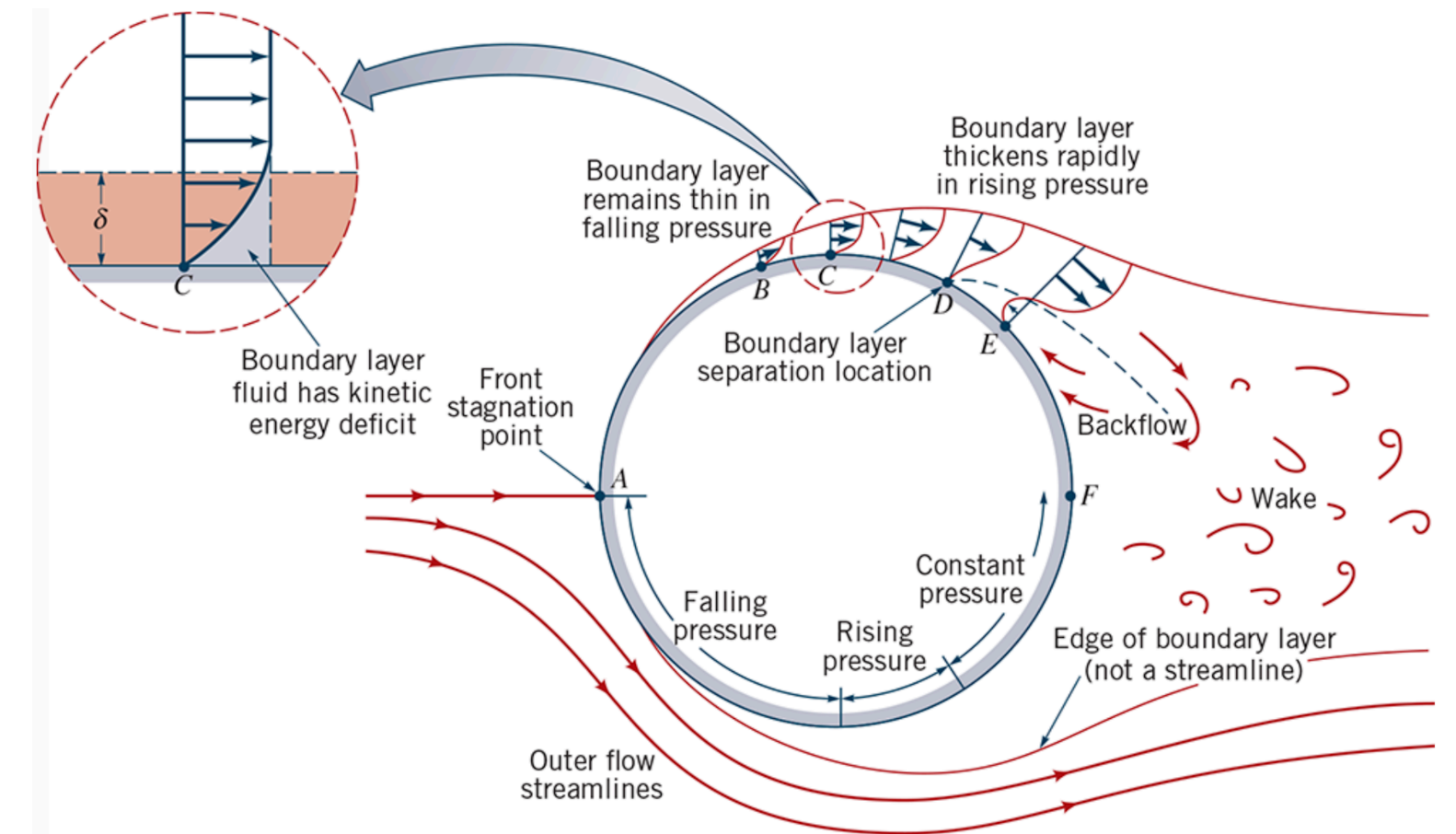
$$\mu = 0 \Rightarrow \mathcal{D} = 0 \quad \text{but} \quad \lim_{\mu \rightarrow 0} \mathcal{D} \neq 0$$

Great mystery in fluid mechanics for more than a century - solved by Prandtl by accounting for viscous effects in thin BL.

Viscous Flow Around a Circular Cylinder and d'Alembert's Paradox

For practical Re , we expect $\delta \ll l_{\text{char}}$ and the BL does not greatly affect the entire flow.

- Therefore, we can reuse the inviscid solution to approximate the wall-tangential and wall-normal pressure gradients in the *viscous* BL.
 - Recall that a fluid particle in the BL experience first a favorable pressure gradient, then an adverse pressure gradient.
 - For the inviscid solution, the *total energy was constant* and the particle balanced its pressure and kinetic energy as it travelled in the BL.
 - However, *in reality, viscous effects dissipate energy* as heat and the particle *cannot overcome the adverse pressure gradient*.
 - The particle slows down (kinetic energy deficit in figure), then stops, and ultimately reverses direction: the BL separates from the surface.
 - A large wake of constant, low pressure is formed behind the separation point, and the drag is dominated by this under-pressure.
 - *Separation causes the drag!*



Delay of Separation from Turbulence

Turbulence can delay separation:

- Later separation point
- Reduced wake size
- Decreased drag

A bowling ball dropped in water illustrates the difference: a smooth surface supports a largely laminar boundary layer, while roughness can trigger turbulence and delay separation.

